

Advanced (Habanero)

Q1. Solve the system: $x + y = 4$, $x - y = 2$

- (A) $x = 3, y = 1$
- (B) $x = 2, y = 2$
- (C) $x = 1, y = 3$
- (D) $x = 4, y = 0$
- (E) $x = 0, y = 4$

Q2. Solve the system: $2x + 3y = 7$, $x - 2y = -3$ using substitution

- (A) $x = 1, y = 2$
- (B) $x = 2, y = 1$
- (C) $x = 3, y = 0$
- (D) $x = 0, y = 3$
- (E) $x = -1, y = -2$

Q3. Solve the system: $x + 2y = 4$, $3x - 2y = 4$ using elimination

- (A) $x = 2, y = 1$
- (B) $x = 1, y = 2$
- (C) $x = 3, y = 0$
- (D) $x = 0, y = 2$
- (E) No solution

Q4. What is the solution to the system: $x - y = 1$, $x + y = 3$?

- (A) $x = 2, y = 1$
- (B) $x = 1, y = 2$
- (C) $x = 3, y = 0$
- (D) $x = 0, y = 3$
- (E) $x = 4, y = 0$

Q5. Solve the system: $2x + y = 4$, $x - 2y = -3$

- (A) $x = 1, y = 2$
- (B) $x = 2, y = 1$
- (C) $x = 3, y = 0$
- (D) $x = 0, y = 4$
- (E) $x = -1, y = -2$

Q6. What is the solution to the system: $x + y = 2$, $2x + 2y = 4$?

- (A) $x = 1, y = 1$
- (B) $x = 2, y = 0$
- (C) $x = 0, y = 2$
- (D) Infinitely many solutions
- (E) No solution

Q7. Solve the system: $x - 2y = -1$, $2x + 3y = 7$

- (A) $x = 1, y = 2$
- (B) $x = 2, y = 1$
- (C) $x = 3, y = 0$
- (D) $x = 0, y = 3$
- (E) $x = -1, y = -2$

Q8. What is the solution to the system: $x + y = 1$, $x + y = 2$?

- (A) $x = 1, y = 1$
- (B) $x = 2, y = 0$
- (C) $x = 0, y = 2$
- (D) Infinitely many solutions
- (E) No solution

Open Ended Questions (FRQ)

Q9. Solve the system: $2x + 3y = 7$, $x - 2y = -3$ using elimination and substitution. Show all work.

Q10. Solve the system: $x + 2y = 4$, $3x - 2y = 4$. Determine if the system has one solution, no solution, or infinitely many solutions. Explain your reasoning.

Chef's Tips

- Read each question carefully
- Use a pencil to fill in the answer bubbles
- Check your work before submitting your test